

30. Answer: D

$$\Delta U = q\Delta V = 2eEx$$

Thus, by conservation of energy,
Loss in KE = gain in PE = $2eEx$

$$KE_f = KE_i + \Delta KE$$

$$\frac{1}{2}mv_f^2 = \frac{1}{2}mv^2 - 2eEx$$

$$\begin{aligned} v_f &= \sqrt{\frac{2\left(\frac{1}{2}mv^2 - 2eEx\right)}{m}} \\ &= \sqrt{v^2 - \frac{4eEx}{m}} \end{aligned}$$

31. Answer: D